

Logic gates problems 2

A	B	C	D	S
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

AB	00	01	11	10
00	1	0	1	1
01	1	0	1	1
11	1	0	0	0
10	1	0	0	0

$$S_2 = A \cdot \bar{C}$$

$$S = (A \cdot \bar{C}) + (\bar{A} \cdot \bar{B})$$

$$S_1 = \bar{A} \cdot \bar{B}$$

$$S = (\bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}) + (\bar{A} \cdot \bar{B} \cdot \bar{C} \cdot D) + (\bar{A} \cdot \bar{B} \cdot C \cdot \bar{D}) + (\bar{A} \cdot \bar{B} \cdot C \cdot D) + (\bar{A} \cdot B \cdot \bar{C} \cdot \bar{D}) + (\bar{A} \cdot B \cdot \bar{C} \cdot D) + (A \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}) + (A \cdot \bar{B} \cdot \bar{C} \cdot D)$$

